

HEALTHPULSE

Smart Personal Health & Fitness Web Application

Your Personal Health & Fitness Companion

1. Problem Statement

- Health information is scattered across multiple applications
- Indian food nutrition data can be difficult to find and track
- Users receive data without meaningful daily insights
- Weight and health history can be difficult to analyze
- Calorie intake and expenditure are not always presented together
- Personalized training and food guidance are often limited or subscription-based

2. Project Objective

- Build an all-in-one personal health platform
- Personalize nutrition and fitness targets
- Make Indian food tracking easier
- Show daily health summaries and long-term progress
- Compare Calories IN vs Calories OUT
- Provide food suggestions and automatic training guidance

3. Personal Health Profile

Users enter:

- Age
- Gender
- Height and weight
- Activity level
- Personal health/fitness goal

The profile powers personalized calculations and recommendations.

4. Health Calculations

HealthPulse calculates:

- BMI — Body Mass Index
- BMR — Basal Metabolic Rate
- TDEE — Total Daily Energy Expenditure
- Daily calorie target
- Protein, carbohydrate and fat targets

Targets are adapted to the user's profile and goal.

5. Smart Indian Food Tracker

Example: User enters “3 dosa”

HealthPulse estimates:

- Calories
- Protein
- Carbohydrates
- Fat

Oil adjustment:

- Less oil
- Moderate oil
- Excessive oil

6. Calories IN, OUT & Daily Summary

CALORIES IN → Food and drinks consumed

CALORIES OUT → Estimated energy expenditure

Daily dashboard can show:

- Calories consumed / target
- Protein, carbs and fat
- Activity
- Goal progress
- Remaining calories

The system turns raw numbers into an easy-to-understand daily summary.

7. Weight History & Goal Tracking

- Record today's and previous weights
- Compare yesterday, weekly and monthly progress
- Display weight trends with charts
- Track personal calorie, macro, activity and weight targets
- Maintain historical food, weight and workout records

8. Food Suggestions & Personal Trainer

Food Suggestions:

- Use remaining calories and nutrition targets
- Consider user's goal and foods already consumed

Automatic Training:

- Uses goal, activity level and training history
- Generates a basic personalized workout
- Includes warm-up, exercises, sets/reps and cool-down
- Can adapt future sessions using progress

9. System Flow & Future Scope

CORE FLOW

Profile → Health Calculations → Food Entry → Nutrition + Oil Adjustment → Calories IN/OUT →
Daily Summary → Suggestions + Training → History & Trends

Future scope:

- AI food image recognition
- Voice food entry
- Barcode scanning
- Water & sleep tracking
- Wearable integration
- Cloud database and user accounts
- Expanded Indian/regional food database